

Sensitivity of Daily Nursing-Home Staffing Assessments to Conserved Reporting Errors

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Working Paper | September 2026

Abstract

Daily nursing home staffing assessments may change when reported hours are assigned to adjacent dates, even if quarterly totals are correct. Using public New York records, we assessed this sensitivity in 589 facilities observed in the first halves of 2024 and 2025. We compared the frequency of below-benchmark days with the depth of joint staffing shortfall, holding facility-quarter totals fixed. Reported incidence declined 1.916 percentage points. Under a 1% registered-nurse hour transfer cap, attained changes ranged from -9.625 to $+5.884$ points. Shortfall declined 1.940 minutes per resident-day, with a conservative change interval of $[-2.578, -1.302]$. A separate 501-facility comparison demonstrated greater variation when the same total allowance could concentrate. These findings show that conserved totals need not support a stable conclusion about daily incidence, whereas shortfall depth can sustain a conditional directional conclusion for research and public reporting. The assumed transfer budget measures neither actual error prevalence nor clinical improvement.

Keywords

nursing-home staffing, administrative data, measurement sensitivity, public reporting, partial information

Introduction

A daily staffing measure and a quarterly staffing total answer different questions. A facility can report the correct number of hours over a quarter while assigning some of those hours to the wrong days. Such errors leave an aggregate accounting check unchanged but can alter the number of days that fall below a staffing benchmark. They can also affect an apparent change over time. The practical question is which conclusions remain supported when the date of a reported hour is less certain than the quarter to which it belongs.

This distinction has a documented reporting mechanism. A federal audit of registered-nurse (RN) staffing submissions identified two sampled items in which overnight hours were assigned to a shift's starting date instead of being divided across calendar days (HHS Office of Inspector General, 2026, p. 9). In one item the monthly total was unchanged. In the other, shifts spanning month boundaries also changed the monthly total. These examples motivate studying calendar allocation, while also identifying a limitation of assuming conservation inside a reporting window. They do not establish the frequency or magnitude of such errors in New York.

The broader importance of staffing measurement is well established. Mukamel et al. (2022) examine daily staffing variation and its association with nursing-home quality. Mukamel et al. (2024) consider incorporating staffing instability into a composite rating. Brunt and Bowblis (2023) relate CMS weekend staffing and turnover measures, alongside a daily-variation measure, to quality indicators. That analysis examines the information in different staffing measures. Brunt and Bowblis (2025) compare alternative measures of daily variation, including their performance against quality outcomes. More recently, Petterson et al. (2026) show that worker-inclusion rules change turnover measurement and facility classifications. Public analyses also already describe New York staffing shortfalls and benchmark-based comparisons (Astrelis, 2026; PBJ320, n.d.). These studies motivate distinguishing changes in the definition of a measure from uncertainty about the records used to calculate it.

New Contribution

This study assesses the sensitivity of public New York nursing-home staffing comparisons to bounded calendar reallocations of RN hours while holding facility-quarter totals and benchmarks fixed. It contributes an empirical distinction between below-benchmark frequency and the depth of joint staffing shortfall: in the same facilities, the direction of the former can reverse under a 1% transfer cap while a conservative bound for the latter remains negative. A separate comparison shows how allowing the same total uncertainty to concentrate changes the assessment. The contribution is conditional measurement evidence from public records, supported by checkable alternative allocations and an explicit distinction between attained values and conservative bounds.

Conceptual Framework

The framework distinguishes staffing provision, the reporting process, and the assessment constructed from a report. Staffing provision concerns who actually worked and when. The reporting process assigns hours to jobs, facilities, and calendar dates. An assessment then combines those assignments with census and a benchmark. A change in an assessment can arise at any of these stages. Our empirical question concerns the last two stages: how much of a conclusion about reported daily staffing depends on the allocation of a fixed number of hours across dates?

Conservation supplies a useful but incomplete accounting restriction. A check of total hours can establish agreement across records while leaving their distribution over days unresolved. A threshold assessment is especially sensitive to that distribution when some days are near a benchmark. Moving hours away from such a day can create another below-benchmark day even if the receiving day remains in its original category. In contrast, a shortfall measure records the amount needed to satisfy the benchmarks. Its aggregate movement is limited by the relocated hours. These are properties of the assessment rules; whether the distinction changes a substantive conclusion is an empirical question.

The allocation of uncertainty across facilities matters separately from its total size. A proportional

cap imposed on every facility-quarter prevents the full allowance from concentrating in facilities whose records can most readily cross thresholds. A pooled cap of the same size removes that restriction. Comparing the two asks what an analyst learns from an assumption about the distribution of possible errors, beyond an assumption about their overall amount. Neither cap describes a measured reporting process in these facilities.

This reasoning yields three questions for public staffing comparisons. Does the direction of an observed period change survive the permitted reallocations? Does the answer differ when the assessment measures shortfall depth rather than below-benchmark frequency? How much does the demonstrated variation increase when the same total allowance can concentrate? For researchers and managers, these questions distinguish a conclusion supported by aggregate accounting from one requiring verification of dates. For policymakers interpreting public reports, they also separate evidence about a named staffing measure from broader claims about compliance or care quality. The analysis evaluates these questions without assigning probabilities to particular errors or estimating a causal policy effect.

Method

Data Sources and Sample

We use the Centers for Medicare & Medicaid Services (CMS) Payroll-Based Journal (PBJ) daily staffing public-use files and the Provider of Services (POS) roster (Centers for Medicare & Medicaid Services, n.d.). The analysis uses reported daily hours for direct RN, licensed practical nurse (LPN), and certified nurse aide (CNA) work, together with the reported daily Minimum Data Set census. The CMS documentation distinguishes the daily facility file from the employee-detail file and separates direct nursing jobs from administrative nursing and director-of-nursing jobs (Centers for Medicare & Medicaid Services, 2023a, 2023b). Our RN, LPN, and CNA measures correspond to job codes 7, 9, and 10 and combine employee and contract hours.

All input hours are retained on the reporting grid of 0.01 hour. Missing, invalid, or unpublished

hours are not set to zero. A finite reported zero is retained as a reported value. A facility-quarter is eligible when its calendar and nursing-hour records are complete and at least 95% of calendar days have a positive integer census. Period analyses require eligibility in every constituent quarter. Among the retained period cohorts, all scored days have valid census. One otherwise observed facility-quarter in 2024Q2 fails the census rule; it is retained in the exception records.

Table 1 distinguishes roster coverage from analytic eligibility. The roster is a quarterly certification snapshot, not proof that a facility operated on every calendar day. Public-file absence can reflect several processes and is not classified as closure, nonreporting, or zero staffing without additional evidence. Public-use-file inclusion also depends on CMS reporting and publication rules (Centers for Medicare & Medicaid Services, 2023a, 2023b). Consequently, the analysis does not represent unobserved facilities automatically.

Table 1. Quarterly Coverage and Period Cohorts.

Quarter	Active POS roster	Observed in PBJ	Eligible facility-quarters
2024Q1	603	601	601
2024Q2	603	597	596
2024Q3	603	598	598
2024Q4	603	596	596
2025Q1	602	596	596
2025Q2	601	594	594
Period cohorts: 2024H1, 595; 2024H2, 596; 2024 full year, 591; 2025H1, 594; matched 2024H1–2025H1, 589.			

Note. POS denotes Provider of Services; PBJ denotes Payroll-Based Journal. A period cohort requires eligibility in all included quarters under a fixed facility identifier. The matched cohort is the intersection of the two first-half cohorts, without inferring corporate continuity from the identifier.

A separate development cohort contains 501 fixed facility identifiers with complete daily records in the first half of 2024: 91,182 facility-days. This cohort was inherited from earlier work requiring availability in subsequent periods and is not a representative sample or a causal survivor stratum. We retain it for the already-computed concentration comparison and report the larger period cohorts separately. All 501 are present in the 2024 period cohorts, while 497 appear in the 2025H1 cohort.

The data checks assess consistency within the public reporting system. In 20 facilities selected

before the expanded linkage exercise, the employee-detail files cover 40 facility-quarters and 355,783 rows. Aggregation to 87,360 facility-day, job, and employment-type values reproduces the daily files without unexplained differences at the centihour level. These files originate from the same reporting process. Their agreement validates linkage and aggregation, not the accuracy of underlying shifts, payroll, or bedside care.

Staffing Benchmarks and Assessment Measures

New York’s June 2023 administrative letter gives numerical references of 3.5 total direct nursing hours, 2.2 CNA hours, and 1.1 RN-plus-LPN hours per resident-day and describes quarterly compliance assessment (New York State Department of Health, 2023). We use these quantities as fixed research benchmarks. Our daily classifications are not official legal determinations. A retained FAQ revised in March 2026 provides additional retrospective guidance, but we do not assume that its full wording or algorithm was communicated in 2024 (New York State Department of Health, 2026).

Let r_{it} , ℓ_{it} , c_{it} denote reported RN, LPN, and CNA *centihours*, and let $N_{it} > 0$ be census for facility i on day t . For an alternative RN record x_{it} , define

$$b_{it}(x) = \mathbf{1}\{x_{it} + \ell_{it} + c_{it} < 350N_{it} \text{ or } c_{it} < 220N_{it} \text{ or } x_{it} + \ell_{it} < 110N_{it}\}. \quad (1)$$

Equality satisfies a benchmark; hours per resident-day are not rounded before comparison. If $\mathcal{T}_{ip}$ is the scored calendar for facility i in period p , the facility score and its equal-facility average are

$$s_{ip}(x) = \frac{\sum_{t \in \mathcal{T}_{ip}} b_{it}(x)}{|\mathcal{T}_{ip}|}, \quad I_p(x) = \frac{1}{n_p} \sum_{i \in \mathcal{I}_p} s_{ip}(x). \quad (2)$$

Here $\mathcal{I}_p$ is the eligible facility cohort and $n_p = |\mathcal{I}_p|$.

Frequency does not measure the hours required to meet the benchmarks jointly. Define joint

shortfall in centihours by

$$d_{it}(x) = \max \{0, 350N_{it} - x_{it} - \ell_{it} - c_{it}, [220N_{it} - c_{it}]_+ + [110N_{it} - x_{it} - \ell_{it}]_+\}. \quad (3)$$

This is the minimum total addition of CNA and licensed nursing hours that would meet all three numerical constraints while holding the existing record fixed. It is a benchmark-completion quantity, not an estimate of clinically necessary staffing or labor demand. Appendix A derives the expression.

Let $V_{ip} = \sum_{t \in \mathcal{T}_{ip}} N_{it}$. Equal-facility shortfall, in minutes per resident-day, is

$$D_p(x) = \frac{0.6}{n_p} \sum_{i \in \mathcal{I}_p} \frac{\sum_{t \in \mathcal{T}_{ip}} d_{it}(x)}{V_{ip}}. \quad (4)$$

The conversion factor is $60/100 = 0.6$. We also report a resident-day-weighted check, replacing the mean of facility ratios by the ratio of the total shortfall to total resident-days. Neither metric is a measure of unobserved delivered care.

Both primary metrics use equal facility weights. Within a facility, incidence counts days, whereas shortfall divides total shortfall hours by resident-days. Matching facility weights therefore removes a between-facility weighting explanation without making the underlying measurement targets identical.

Calendar-Allocation Sensitivity Analysis

The framework is deterministic and conditional on the retained records; it is an empirical application rather than a new general optimization or inference theorem. Its intervals are not confidence intervals, its budgets are not estimated error rates, and its alternative records are not reconstructed payroll truth.

Permitted Records and Transfer Budgets

Within each facility-quarter, let f_{ist} be nonnegative integer centihours moved from reported day s to adjacent calendar day t , with $s \neq t$. We require

$$f_{ist} = 0 \quad \text{unless } |s - t| = 1 \text{ and } s, t \text{ are in the same quarter,} \quad (5)$$

$$\sum_t f_{ist} \leq r_{is}, \quad (6)$$

$$x_{it} = r_{it} - \sum_u f_{itu} + \sum_s f_{ist}, \quad (7)$$

$$\sum_{s,t \in q} f_{ist} \leq B_{iq}(\varepsilon) = \left\lfloor \varepsilon \sum_{t \in q} r_{it} \right\rfloor. \quad (8)$$

The supply restriction uses the original record. Incoming hours do not expand a day's transferable supply; the model therefore does not relay a received hour along several days. Hours remain in the same facility. Transfers preserve RN totals within each closed quarter, and LPN, CNA, and census values stay fixed.

We evaluate the existing grid $\varepsilon \in \{0, 0.005, 0.01, 0.02, 0.05, 0.10\}$. A 1% cap concerns the mass of RN hours that may be relocated, not the percentage of facilities, employees, or days with errors. The amount moved is the sum of transfer flows; it is distinct from the sum of absolute changes in final daily hours. Opposing changes can offset in the latter accounting quantity.

The audit evidence supports the existence of calendar-allocation errors. It does not establish that errors are universally adjacent, RN-only, bidirectional, proportional to each facility's total hours, or confined to a quarter. These are restrictions defining the sensitivity exercise. In particular, the nationwide net unsupported-hours estimate in HHS Office of Inspector General (2026) does not calibrate this gross date-transfer budget. The primary set allows both directions; an audit example of start-date reporting does not establish that all facilities use that process.

The quarter closure is consequential. With fixed census, within-quarter hour conservation leaves ratio-of-sums staffing proxies unchanged, but not necessarily a daily classification or an official regulatory calculation. We retain separate boundary and one-direction comparisons from

the underlying analysis, rather than treating conservation at the edge of the sample as an observed fact. Appendix D describes their interpretation.

Attainment, Outer Bounds, and Period Contrasts

For a fixed cohort, we minimize and maximize $I_p(x)$ over the permitted records. Under separate facility-quarter budgets the feasible set is a product across those units. Saved alternative records are reconstructed and checked for integer flows, original supplies, support, budgets, conservation, and the resulting score. A solver status alone does not establish validity.

We distinguish three statements. A feasible record establishes an attained value. An attained value that matches a valid optimization bound establishes an extreme. An outer bound contains all permissible values but can be conservative. Integer scores need not fill every point between two extremes. All statements of exact endpoints in this paper mean that the attained integer counts match the retained numerical optimization or analytic certificates after independent record checks; they are not claims of a formally verified exact-arithmetic solver.

For the matched facilities, write $p = 0$ for 2024H1 and $p = 1$ for 2025H1. There is no restriction that the error pattern be identical across years. With disjoint periods and their separately imposed budgets, the extreme changes in incidence are

$$\underline{\Delta I} = \underline{I}_1 - \bar{I}_0, \quad \overline{\Delta I} = \bar{I}_1 - \underline{I}_0. \quad (9)$$

Combining the corresponding records is feasible across the two periods. This interpretation would change under an additional common-error-pattern restriction.

For shortfall, we use a transparent outer bound. Holding ℓ, c, N fixed, $d_{it}(x)$ is nonincreasing and 1-Lipschitz in RN centihours. Moving at most B centihours can increase total shortfall by at most B and decrease it by at most B . Thus

$$|D_p(x) - D_p(r)| \leq \rho_p(\varepsilon) = \frac{0.6}{n_p} \sum_{i \in \mathcal{I}_p} \frac{\sum_{q \subset p} B_{iq}(\varepsilon)}{V_{ip}}, \quad (10)$$

and a valid interval for the matched change is

$$\Delta D(r) \pm \{\rho_0(\varepsilon) + \rho_1(\varepsilon)\}. \quad (11)$$

Appendix B supplies the elementary argument. A negative upper bound certifies a decrease throughout the error set. A bound that crosses zero does not establish that an increase can be attained.

Concentrating a Fixed Total Budget

The separate caps in equation (8) limit how much error can be concentrated. In the 501-facility development cohort we compare them with a shared cap equal to their sum,

$$\sum_{i,q} \sum_{s,t \in q} f_{ist} \leq B^{\text{sum}}(\varepsilon) = \sum_{i,q} B_{iq}(\varepsilon), \quad (12)$$

while retaining within-facility, within-quarter support and original supplies. The shared constraint replaces the separate caps. It permits budget concentration; it does not permit transferring hours between facilities.

Earlier pooled computations used a single floor after summing hours. For the comparison here, we verify that both retained pooled witness records also fit the smaller sum-of-floors cap in equation (12). We report their attained values without claiming that they exhaust the pooled possibilities. The mathematical inclusion of the separately capped set is immediate; the empirical question is how much additional variation is demonstrated by the records.

Results

Changes in the Same Facilities

Table 2 presents the central comparison. In the same 589 facilities, the reported incidence change is -1.916 percentage points. At a 1% cap, both an additional decrease and an increase are attainable:

the extreme changes are -9.625 and $+5.884$ percentage points. Therefore the sign of the reported incidence decline is not determined throughout this error set.

Joint shortfall, using equal facility weights, falls from 22.263 to 20.323 minutes per resident-day, a change of -1.940 . Its outer interval at 1% is $[-2.578, -1.302]$. Because even the upper bound is negative, the decline in this particular shortfall measure persists in every permitted record. The resident-day-weighted check also certifies a decline. This is a distinction between assessment targets, not evidence that one target is universally more accurate or clinically preferable.

Table 2. Matched Change From 2024H1 to 2025H1 Under a 1% RN Transfer Cap.

Target and weighting	Reported change	Sensitivity limits	Status
Below-benchmark days, equal facility weights (percentage points)	-1.916	$[-9.625, 5.884]$	Both extremes attained
Joint shortfall, equal facility weights (minutes/resident-day)	-1.940	$[-2.578, -1.302]$	Outer bound
Joint shortfall, resident-day weights (minutes/resident-day)	-2.207	$[-2.815, -1.598]$	Outer bound

Note. The cohort is fixed at 589 facilities in both periods. Each facility-quarter has its own cap, based on its reported RN hours. Error patterns may differ across years. The shortfall limits are not certified attainable. Limits concern fixed records and assumed errors, not sampling uncertainty.

Figure 1 retains every previously evaluated budget point. Incidence changes of both signs are already feasible at the smallest positive grid point, 0.5%. The shortfall decline remains certified at 2%, with an upper bound of -0.664 minutes per resident-day. At 5% the outer bound crosses zero. That crossing marks loss of this conservative certificate, not an observed or attained reversal in shortfall. At 10%, small optimization gaps remain for incidence; the attainable spans and outer bounds are reported separately in Appendix Table 5.

Error Concentration and Separate Caps

In the separate 501-facility cohort, reported incidence is 69.358% in 2024H1. At 1%, separate facility-quarter caps give extremes of 66.047% and 73.394%, a width of 7.347 percentage points. The pooled witness records produce 64.429% and 76.170%. Their span is 11.740 percentage points, exceeding the entire separately capped range (Table 3).

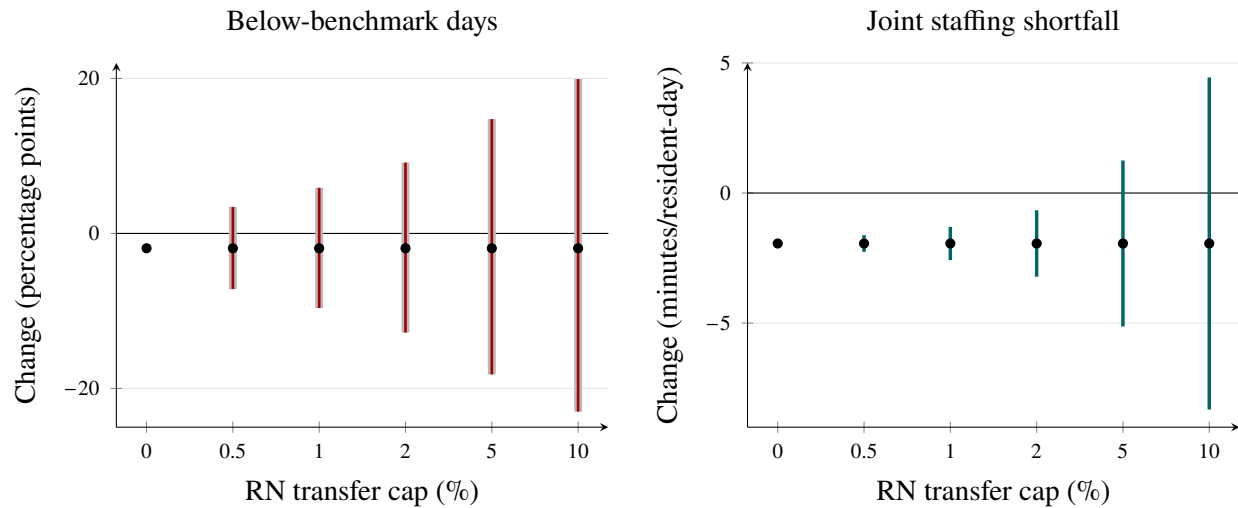


Figure 1. Sensitivity of Staffing Changes to the RN Transfer Cap in 589 Matched Facilities.

Note. Both metrics use equal facility weights. Black points are reported changes. Left: red lines join attained endpoints; gray lines show certified outer bounds. Right: teal lines show conservative outer bounds whose endpoints need not be attainable. The very small 10% incidence gaps are not visually separable at this scale. Budget points are equally spaced categories, with no interpolation or sampling-confidence interpretation.

Table 3. Concentration Comparison With the Same Total 1% Budget in 501 Facilities.

Budget restriction	Lower (%)	Upper (%)	Span (pp)	Status
Separate facility-quarter caps	66.047	73.394	7.347	Both extremes certified
Shared cap, concentration allowed	64.429	76.170	11.740	Two attained values; span is a lower bound

Note. The sum of the separate caps is 7,597,424 centihours. The pooled lower-value and upper-value witnesses use 7,591,666 and 7,596,515 centihours, respectively. Both fit that same cap. All 501 facilities have 182 valid days, so equal facility weighting equals equal facility-day weighting. The shared-cap extremes have not been certified.

This is not an integer-rounding artifact. The earlier pooled cap, formed by flooring once after summing all reported RN hours, exceeded the sum of individual floors by 446 centihours. Both witness records fit the smaller cap. The comparison shows that the information content of the primary range depends on limiting error concentration, even when the total permitted amount is held fixed.

Levels Across Larger Period Cohorts

Table 4 gives the corresponding incidence levels in the larger eligible cohorts. The 1% widths range from 7.454 to 8.007 percentage points. Annual aggregation does not make the sensitivity disappear because the error budget remains available within every quarter; the design does not impose cancellation across quarters.

Table 4. Incidence Levels Under Separate 1% Facility-Quarter Caps.

Period	Facilities	Reported (%)	Min.–max. (%)	Width (pp)
2024H1	595	68.059	64.749–72.203	7.454
2024H2	596	67.125	63.611–71.435	7.824
2024 full year	591	67.654	64.233–71.891	7.658
2025H1	594	66.579	63.031–71.038	8.007

Note. All displayed endpoints match the retained bounds. Period cohorts differ; subtracting rows is not the matched 589-facility comparison in Table 2. The benchmark, job mapping, and separate quarterly budgets remain fixed.

The larger 2024H1 cohort also clarifies the development sample’s limits. The 94 added facilities have lower reported incidence than the original 501, 61.14% versus 69.36%, while their 1% range width is 8.03 rather than 7.35 percentage points. The expansion is informative about observed sample composition, not proof of representation of the missing facilities.

Additional Findings

Two findings constrain the interpretation of computational and ranking results. First, a simple analytic count bound is already close to the fully certified mean range. In the 501-facility cohort at 1%, its width is 7.3995 percentage points, only 0.0526 percentage points wider than 7.3468. The aggregate result therefore does not establish a large practical advantage from sophisticated

optimization alone.

Second, the binary score has a pronounced ceiling. In the four larger period cohorts, 91–117 facilities have a reported score of one, while a hypothetical top-10% selection contains 60 facilities. Under a fixed ascending facility-identifier tie rule, the 1% transfer scenarios can replace 20–23 of those 60 selections. Changing tie order alone is a different exercise and permits 31–57 replacements (Appendix Table 6). These quantities are not additive. We retain ranking as a secondary diagnostic and do not interpret the tie rule as a clinically validated inspection priority.

Some apparent stability also follows from holding other staffing channels fixed. In the 501-facility cohort, 55,083 of 91,182 days, or 60.41%, are below the CNA benchmark. Those days remain below at least one benchmark regardless of RN reallocation. Descriptive sensitivity widths are larger in some facilities with greater contract-staffing shares, but this neither identifies their actual reporting-error rates nor a causal effect of contract staffing.

Job labels matter as well. One facility reports zero direct-RN hours in all six examined quarters while reporting positive director-of-nursing and administrative-RN hours. Its CNA benchmark is also below the floor on every observed day. We do not describe it as having no registered nurses or as misreporting. Its invariance illustrates both a zero proportional transfer budget and an independently fixed CNA channel. No validated gain in quality prediction or inspection outcomes has been established by the preceding exploratory work, and no such gain is claimed here.

Discussion

The analysis identifies a difference in the strength of two statements about observed staffing records. A fall in the reported below-benchmark day share is a correct description of the data, yet its direction can reverse within the specified calendar-allocation set. The decline in joint shortfall survives the same 1% cap under a conservative bound. Neither statement establishes how actual staffing or resident care changed. Keeping these levels of interpretation separate makes the sensitivity exercise useful without turning it into an error-correction estimator.

The two targets respond differently to the location of a moved hour. Incidence changes only

when a benchmark boundary is crossed. A small transfer can therefore change the classification of a day just above a boundary while leaving the receiving day's classification unchanged. Shortfall instead records the size of the remaining deficit. Its aggregate change is limited by the amount moved even when several classifications change. This mechanism explains why the two sensitivity calculations need not agree; it does not by itself determine their empirical magnitudes. The retained records establish the incidence reversals, and the bound establishes shortfall's directional stability. Differences in within-facility day normalization remain part of the respective targets.

For users of daily staffing information, the exercise indicates which conclusion needs date-level verification to be sustained under a given error allowance. Agreement on quarterly totals would not resolve the incidence ambiguity demonstrated here. Conversely, if an analyst is willing to impose the stated 1% or 2% restrictions, additional optimization of shortfall is not necessary to certify the sign of its change. These are conditional statements about verification needs, not estimates of the returns to auditing or recommendations to replace a clinical quality measure.

An analyst communicating this comparison could report the observed change, the permitted transfer cap, and whether each conclusion survives that cap. For the matched comparison, this would qualify the incidence decline while retaining a conditional statement about shortfall decline. The result would not establish regulatory compliance or rank facilities for intervention. Likewise, a bound crossing zero would call for an inconclusive statement unless an opposite-sign record had actually been constructed. This reporting distinction is available from the existing public inputs.

The practical implication is to align a verification task with the conclusion being communicated. If a manager needs to defend a decline in the number of below-benchmark days, reconciling total quarterly RN hours alone is insufficient under the scenarios examined here. Checking how shifts were allocated across midnight addresses the uncertainty more directly. If the question instead concerns the aggregate depth of benchmark shortfall, the reported decline survives the stated small transfer allowances without reconstructing every correct date. This difference does not identify an optimal audit strategy or justify reducing staffing oversight; it identifies the information missing from a particular claim.

The concentration result adds a qualification for system-level reporting. A reassuring assessment based on separate facility caps can depend on assuming that no facility contributes disproportionately to the total uncertainty. Reporting the pooled witnesses makes that assumption visible. A broader application would need to justify both the size and the distribution of its error allowance. The New York estimates cannot supply those justifications for another reporting system, but the distinction between the two assumptions remains relevant when evaluating its data.

Limitations

The most important limitation is the unknown error budget. Public PBJ data lack clock-in and clock-out times, payroll validation, and independent observations of the correct allocation. The federal audit supplies an example of the mechanism but not a New York calibration. Reporting a grid exposes the assumption; it does not make any grid value empirically correct. An error magnitude or a common-year software pattern justified by independent evidence could alter the assessment, and the present results do not rank such patterns by likelihood.

The error set is also deliberately partial. It isolates RN dates while holding CNA, LPN, census, and total RN hours fixed inside quarters. Errors in those other quantities, longer displacement, additions and deletions, or boundary flows can invalidate a conclusion stated without its restrictions. In particular, stability produced by the fixed CNA floor is not robustness to all nursing-data errors. The concentration comparison further shows that even the same total transfer allowance can imply different information when its allocation is restricted differently.

These limits coexist with an empirical use: the procedure audits a clearly defined comparison using public inputs and checkable alternative records. It separates a feasible contradiction from an inconclusive outer bound, reports the assumptions that support each conclusion, and demonstrates that a general statement of staffing improvement is less precise than an explicit statement about a named measure. This use does not require inventing a new general theorem or claiming that the measures capture every policy objective.

The numerical intervals describe the retained finite facility cohorts, so their validity does not

rely on treating facilities as a random sample. Generalization to other states, missing facilities, or future reporting systems would require additional evidence about threshold proximity, staffing composition, and error restrictions. Sampling uncertainty for such a broader target is outside the present calculation.

Finally, the analysis is retrospective. The development cohort and earlier descriptive results were observed before the general calendar-transfer design was fixed. Staffing and quality information for 2025 had also been examined previously. The matched-period diagnostic was pursued after observing a decline. The later period is therefore a same-design extension with acknowledged exposure, not a pristine holdout or a preregistered test. We preserve the full evaluated budget grid, unchanged sample rules, adverse findings, and unresolved high-budget bounds. The contribution remains conditional measurement evidence in the observed eligible facilities.

Conclusion

Conserving facility-quarter RN totals does not ensure that a daily staffing assessment, or its change over time, is stable to calendar allocation. In matched New York facilities, the direction of a below-benchmark incidence change can vary within a 1% transfer set even while the decline in a joint shortfall measure is certified. Permitting concentration of the same total transfer allowance further enlarges demonstrated aggregate variation. These results support reporting the assessment target, the permitted recording errors, and the status of the resulting bounds together.

Appendix A. Joint Benchmark Shortfall

Suppress facility and day indices. Let $a_C, a_L \geq 0$ be additional CNA and licensed-nursing centihours, respectively, where licensed additions may be RN or LPN. Joint benchmark completion requires

$$a_C \geq [220N - c]_+, \quad a_L \geq [110N - x - \ell]_+, \quad a_C + a_L \geq 350N - x - \ell - c.$$

The minimum of $a_C + a_L$ is equation (3): the sum must exceed both the total deficit and the sum of the two group deficits. Starting with the two group minima and allocating any remaining total deficit to either group attains their maximum. All inputs and constraints are on the centihour grid. No wage, substitution preference, or resident outcome enters this accounting quantity.

Appendix B. Why the Continuous Bound Uses Moved Mass

Each term inside the maximum in equation (3) is nonincreasing and 1-Lipschitz in x ; the same is true of the maximum. For a permitted final allocation write $\delta_t = x_t - r_t$. Let $P = \{t : \delta_t > 0\}$ and $M = \{t : \delta_t < 0\}$. Monotonicity implies that increases in shortfall can occur only on M and decreases only on P . Hence

$$-\sum_{t \in P} \delta_t \leq \sum_t \{d_t(x_t) - d_t(r_t)\} \leq \sum_{t \in M} (-\delta_t).$$

From the transfer reconstruction, each sum on the left or right is no greater than $\sum_{s,t} f_{st} \leq B$. Thus the absolute change in unweighted total shortfall is at most B , not merely $2B$. Summing quarter bounds, dividing by each facility's fixed period resident-days, and averaging gives equation (10). Summing the bounds from the two disjoint periods gives equation (11). For resident-day weights the corresponding period radius is

$$\rho_p^{RD}(\varepsilon) = 0.6 \frac{\sum_{i,q \subset p} B_{iq}(\varepsilon)}{\sum_i V_{ip}}.$$

This elementary bound does not require its endpoints to be attainable under adjacency or original-supply restrictions. Those restrictions can only make it conservative. It concerns the defined aggregate shortfall, not arbitrary alternative weights on each day.

Appendix C. Full Grid for the Matched Comparison

Table 5. Matched Change at Every Evaluated Transfer Cap.

Cap (%)	Incidence change (percentage points)				Shortfall change (minutes/resident-day)	
	Outer lower	Attained lower	Attained upper	Outer upper	Outer lower	Outer upper
0	-1.916	-1.916	-1.916	-1.916	-1.940	-1.940
0.5	-7.184	-7.184	3.425	3.425	-2.259	-1.621
1	-9.625	-9.625	5.884	5.884	-2.578	-1.302
2	-12.791	-12.791	9.143	9.143	-3.217	-0.664
5	-18.196	-18.196	14.755	14.755	-5.131	1.250
10	-23.004	-22.998	19.916	19.928	-8.321	4.441

Note. Both outcomes use the same 589 facilities and equal facility weights. Shortfall outer endpoints are not claimed attainable. The reported incidence and shortfall changes are constant across rows; only the permitted records change. The 10% incidence upper-endpoint gap is 0.011256 percentage points before rounding; it is preserved rather than reported as exact. Zero-cap identity does not imply accuracy of the underlying record.

Appendix D. Rank Ties and Other Scope Checks

Table 6. Secondary Top-10% Diagnostic and Separate Tie-Order Sensitivity.

Period	Selected	Tied at score 1	Max. replacements at 1% cap	Tie-only max. replacements
2024H1	60	117	20	57
2024H2	60	117	23	57
2024 full year	60	91	21	31
2025H1	60	117	23	57

Note. The third column counts the ceiling tie that contains the selection cutoff. Transfer-induced replacements use a fixed ascending facility identifier for ties and jointly feasible records. The last column instead changes the order within the original score ties, holding the staffing record fixed. These are distinct comparisons and cannot be added.

The retained boundary comparison joins the first half of 2024 to one outside day at each end and allows the internal quarter-boundary edge. Any transfer incident to a target quarter consumes that quarter's budget; an internal cross-quarter transfer consumes both budgets but is counted once as physical moved mass. This prevents a free import from an outside day. Conservation then holds for the padded window rather than necessarily for the target quarter or half-year. The comparison changes boundary bookkeeping and must not be interpreted as the primary closed-quarter set.

A separate start-day-to-next-day scenario restricts direction. An additions-and-deletions scenario instead bounds the absolute amount of changed RN hours and does not use the moved-mass budget.

Equal nominal percentages do not make these different budgets interchangeable. Same-day RN–LPN relabeling preserves licensed and total direct-nursing hours, so by itself it leaves equations (1) and (3) unchanged with CNA fixed. This invariance is an accounting check, not evidence that role reporting is accurate.

The largest period-specific 1% widths in the descriptive contract-staffing comparison occur in the above-25% contract-share group: 10.13, 10.30, 10.24, and 10.27 percentage points across 2024H1, 2024H2, 2024 full year, and 2025H1, respectively. The zero-contract-share group has widths 4.87, 5.33, 4.90, and 5.67. Group definitions use the original reported employment mix. The comparison does not hold all facility characteristics constant, identify an error process, or establish causality.

Appendix E. Sources, Computation, and Reproducibility

The empirical literature comparison uses the retained primary texts of Mukamel et al. (2022, 2024) and Petterson et al. (2026) and the primary abstracts and bibliographic metadata of Brunt and Bowblis (2023, 2025); access to their full articles was unavailable. Claims about the absence of analyses in their unread sections are not made. The producer pages (Astrelis, 2026; PBJ320, n.d.) establish that related public analyses exist; their numerical, clinical, and legal conclusions have not been independently validated here. This is a targeted comparison, not a systematic review establishing first use.

The federal audit’s calendar examples motivate an error mechanism, not every restriction in the model. A separately inspected Texas facility audit (Texas Health and Human Services Office of Inspector General, 2022) contained a management response describing midnight splitting; it was not adopted as a second confirmed instance of an uncorrected calendar-allocation error. The source-access and rejected-lead records are retained with the project.

The computation uses integer transfer records, mixed-integer optimization for daily classifications, analytic bounds, and independent reconstruction checks. The development implementation was checked against exhaustive small allocation problems and zero-budget and conservation identities.

Unresolved numerical bounds are retained in the result tables. The simpler continuous bound in Appendix B is used for the period-change claim without requiring a new shortfall optimization.

The manuscript uses existing saved empirical outputs; it introduces no additional estimation. The companion archive links the main results to frozen public facility-day inputs, integer transfer records, and the original result tables. Its verification script uses the Python standard library and runs independently of the development project. Instructions are provided in the accompanying documentation.

The verifier reconstructs the separate-cap 1% incidence endpoints across the four period cohorts, the matched-period comparison, and the two concentrated-budget records. It also recalculates the matched shortfall bounds at every reported budget and checks the figure against the saved tables. It does not rerun optimization or independently prove solver dual validity. Incidence endpoints at other budgets are retained as saved results; their full transfer records remain in the research project.

The companion documentation describes the inputs, expected outputs, source provenance, and scope of the verification, together with instructions for rebuilding the manuscript. The archive contains facility-level public inputs and research outputs; employee-detail files and third-party full texts are not included.

Replaying the saved inputs is distinct from downloading the evolving CMS files again or validating original payroll truth. No new survey, confidential payroll file, or private employee linkage is needed for these conditional computations. The accompanying artifact inventory records what is included and what remains in the development archive.

Use of Generative Artificial Intelligence

OpenAI Codex assisted with literature searches, research planning, code generation and debugging, execution of computational checks, preparation of data-based figures, and drafting and revision of the manuscript and submission materials. Numerical findings were computed from public inputs using executable code; they were not elicited as numerical predictions from a language model. The companion provides saved records and reconstruction checks. This disclosure covers assistance

beyond language editing. Responsibility for the submitted analysis, source interpretation, and final text rests with the human author.

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